

Ideation Asset

1. Problem Statement

The process of coordinating group travel is fundamentally inefficient, friction-prone, and logistically fragmented. Group organizers currently face significant friction when trying to manually reconcile disparate financial constraints, divergent personal interests and varying itinerary expectations.

This operational complexity is compounded by three critical systemic vulnerabilities:

- **The Failure of Democratic Decision-Making:** Relying on traditional majority voting structures routinely marginalizes minority preferences, leading to inequitable itinerary outcomes and decreased user satisfaction.
- **Compromised Risk Management:** Critical individual constraints, including severe medical allergies, strict dietary restrictions, and physical mobility limitations are frequently obscured within decentralized, chaotic communication channels, introducing logistical and safety risks.
- **High Administrative Overhead:** Due to the absence of a unified system, coordinators must bear a severe administrative burden, manually aggregating data, tracking consensus and scheduling timelines across siloed spreadsheets and disconnected messaging applications.

Consequently, there is a clear market need for an automated, centralized orchestration system that optimizes group alignment while mitigating manual coordination overhead.

2. Project Objectives

The primary objective of this initiative is to architect and deploy an AI-powered mobile application designed to automate and optimize group travel orchestration. The platform aims to eliminate legacy manual coordination by deploying three core algorithmic capabilities:

- **Algorithmic Preference Harmonization:** Utilizing machine learning models to dynamically synthesize and reconcile disparate budgets, timelines, and recreational interests into optimized travel options.

- **Consensus-Driven Decision Logic:** Replacing traditional majority-vote mechanisms with a weighted consensus engine that systematically protects and integrates minority user preferences.
- **Automated Risk and Constraint Mitigation:** Embedding a hard-constraint verification layer that cross-references all recommended itineraries against user-defined medical, dietary, and physical limitations.

By consolidating communication, scheduling and validation into a unified, cloud-synchronized interface, the platform drastically mitigates administrative overhead and ensures inclusive trip planning.

3. Target User Persona Segmentation

To ensure precise UX alignment, the ideation phase established two distinct primary user vectors:

- **The Lead Facilitator ("The Coordinator"):** Responsible for operational logistics, vulnerable to administrative burnout, and burdened with manually reconciling data across decentralized channels.
- **The Constrained Participant ("The Niche Traveler"):** Possesses strict, non-negotiable financial, dietary, or mobility profiles, and is statistically prone to systemic marginalization under standard democratic voting rules.

4. Functional Value Mapping

Operational Inefficiency	Algorithmic Solution	System Architecture Function
Financial & Interest Disparity	Multi-Variable Data Synthesis	Preference Harmonization: Generates optimized activity nodes within overlapping budget curves.
Minority Voice Marginalization	Non-Zero-Sum Voting Models	Consensus Weighting: Factors individual "Must-Do" priorities directly into the core generation logic.
Oversight of Health Constraints	Deterministic Exclusion Filtering	Automated Guardrails: Programmatically blocks venues violating medical or mobility restrictions.
Fragmented Tool Architecture	Centralized Lifecycle Management	Unified Workspace: Consolidated dashboard for profile building, voting, and dynamic scheduling.

5. Product Feature Roadmap (MoSCoW Framework)

Core Capabilities

- Asynchronous multi-user onboarding via secure, tokenized invitation protocols.
- Granular individual profile configuration tracking fiscal limits, interest vectors, and health constraints.
- High-priority "Must-Do" activity preservation mechanisms.
- Real-time consensus-based voting modules.

Enhanced Logic

- Automated system flags that trigger if manual overrides breach a user's health constraint.
- Dynamic timeline workspace featuring concurrent user commenting and micro-feedback threads.
- Automated contextual substitution engine provides real-time activity alternatives upon node rejection.

Scale Features

- Direct Merchant Integration: Global Distribution System (GDS) integrations for in-app accommodation, transit, and ticketing checkout.
- Geographical Discovery Hub: Regional state-by-state explore engine categorized by distinct travel modalities (*Nature, Adventure, Leisure*).
- Social-Financial Utilities: Encrypted messaging components, integrated ledger expense tracking, and real-time intra-trip location telemetry.

Appendices

Brain Dump

CodeNecton 2026

Daniel Koh Yu Hang • alyapip's Screen

720p 30FPS LIVE

CodeNecton Malwaffle_260905_225504.pdf

Individual (plan travel)

Budget
makan
aktivite
transport
hotel

AI suggestion
Place/activ
food

user choose
which
activity

Itinerary
Suggestion
Add/remove
ent Itinerary

Group

Budget
min max
Voting budget

Average
Budget
Place
Suggestion
food

Itinerary
AI suggest itinerary
add/remove

allyapip

allyapip

rii

Aly4mht

alyapip

daniao

diners

LIVE

SKILL ISSUE

allyapip

rii

Aly4mht

alyapip

daniao

diners

allyapip

rii

Aly4mht

alyapip

daniao

diners